TSM 098

# Homework 3

Combining terms, distribution, and polynomial products

**Topics 9-12** | Estimated time: about 1 hour

**Complete all problems and show your work on separate paper.** Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

## Terms and factors

**1.** Simplify.

$5x - 3x + 7 - 10$

**2.** Simplify.

$4a^2 + 3a - 2a^2 + 5a$

**3.** Simplify.

$(-3x^2)(4x^3)$

**4.** Simplify.

$2a \cdot 3a \cdot a^2$

**5.** Simplify.

$(2x^2)^3$

## Read the operation

**6.** Simplify 3x + 3x and (3x)(3x). Explain why only the product changes the exponent.

## Distribution and sums

**7.** Expand as needed and combine like terms.

$4(2x - 3) - 3(x + 2)$

**8.** Expand as needed and combine like terms.

$-2x(3x^2 - 4x + 5)$

**9.** Expand as needed and combine like terms.

$(3x^2 - 5x + 2) + (2x^2 + x - 7)$

**10.** Expand as needed and combine like terms.

$(5x^2 + 2x - 4) - (2x^2 - 3x + 1)$

**11.** Expand as needed and combine like terms.

$2(x-3)+4(2x+1)-5x$

## Polynomial multiplication

**12.** Multiply and combine like terms. Show every product.

$(x+3)(x-5)$

**13.** Multiply and combine like terms. Show every product.

$(2x-1)(3x+4)$

**14.** Multiply and combine like terms. Show every product.

$(x+4)^2$

**15.** Multiply and combine like terms. Show every product.

$(x-2)(x^2+3x+1)$

## Larger products and checking

**16.** Multiply and combine like terms.

$(x^2+x+1)(x^2-x+2)$

**17.** Check your answer to Problem 16 by evaluating both the original product and your expanded polynomial at x = 1. Explain why this check alone is not a proof for every x.